The Hillingdon Hospitals
NHS Foundation Trust

Sepsis in Maternity

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Guideline Number	567
Version Number	5.0
Date Issued	2nd July 2025
Next Version Date	2nd July 2028
Approval Group	Maternity Document Review Group
Approval Date	11th June 2025
Ratified Group	Clinical Outcomes and Effectiveness Group
Ratified Date	2nd July 2025

Change History

Version	Author	Summary of changes
4.0	Claire Chambers, Obstetrics and Gynaecology Registrar	Full review with updates from the latest UK Sepsis Trust manual
5.0	Taruna Agrawal, Clinical Fellow for Maternal Medicine & Risk Sarah-Jane Lam, Consultant Obstetrician	Full review

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1. Aim/Purpose

This guideline covers:

- Detection and management of suspected sepsis in pregnancy for antenatal, intrapartum and postnatal patients
- Investigations
- Sepsis prevention

2. Introduction

Sepsis is one of the leading causes of maternal morbidity and mortality in the UK. In the Mothers and Babies Reducing Risk through Audits and Confidential Enquiries across the UK (MBRRACE-UK) (2024) report sepsis was responsible for 9% of all maternal deaths. Sepsis was the fifth leading cause of direct and indirect maternal deaths causing 25 maternal deaths in the 2020-2022 triennium. Thirty-eight of the deaths that occurred between March 2020 and December 2022 were directly attributed to COVID-19 infection. COVID-19 remains the leading cause of indirect maternal death in 2020-22 with a maternal mortality rate of 1.98 per 100,000 maternities (95% CI 1.40-2.72) based on the number of maternities between March 2020 and December 2022, the period of the pandemic. The National Maternity and Perinatal Audit (NMPA) report examining maternity admissions to intensive care in 2015/2016 highlighted the importance of infection as the second most common cause for admission, after haemorrhage. The most common infections were pneumonia (44%), urinary tract (20.4%) and genital tract infections (18.5%).

To avoid preventable deaths the importance of maternal vaccination for influenza and COVID-19 must be continually promoted.

Sepsis in pregnancy and the puerperium can be difficult to diagnose and treat. The immunological adaptations in pregnancy make patients more susceptible to infection and prone to rapid progression to septic shock. Pregnant women are generally young and fit, so can compensate for the physiological changes that occur in sepsis, appearing deceptively well often up to the stage of collapse. Early recognition and accurate diagnosis, together with aggressive treatment strategies, are essential to prevent maternal morbidity and mortality from sepsis. Midwives, doctors and other health professionals need to “think sepsis” and implement ‘The Sepsis 6’ bundle, including giving antibiotics within an hour of the diagnosis being suspected.

Sepsis Definitions

- **Sepsis** is defined as a life-threatening organ dysfunction due to a dysregulated host response to infection.
- **Septic shock** is defined as a subset of sepsis where particularly profound circulatory, cellular and metabolic abnormalities substantially increase mortality. Patients with septic shock include those who require vasopressors to maintain a mean arterial pressure of at least 65mmHg and who have a serum lactate of at least 2mmol/L despite fluid resuscitation.
- Pregnant women are 20 times more likely to develop invasive Group A beta-haemolytic Streptococcus (iGAS) infection compared to non-pregnant women, and are 80 times more likely up to 28 days postpartum.

3. The Guidance.

3.1 Risk Factors for developing sepsis during pregnancy/postpartum

Increased susceptibility to infection

- Obesity
- Impaired glucose tolerance / diabetes or other comorbidities
- Impaired immunity (e.g. HIV, or on immunosuppressant medication)
- Anaemia
- Black or minority ethnic group origin
- Maternal age>35 years
- Intravenous drug use

Infection history / carrier

- Vaginal discharge
- Prolonged spontaneous rupture of membranes (SROM) or preterm pre-labour rupture of membranes (PPROM)
- History of pelvic infection
- History of Group B streptococcal infection
- Group A ß haemolytic Streptococcus / other invasive organisms - acquisition / carriage
- Young children at home

Invasive genital tract procedure

- Amniocentesis and other invasive procedures
- Cervical cerclage
- Vaginal trauma
- All forms of operative vaginal delivery
- Complication of caesarean section
- Manual removal of placenta
- Wound haematoma
- Retained products of conception

3.2 Sources of infection

Source	Symptoms/Signs	Diagnoses to consider
Central Nervous System	Headache, neck pain/stiffness, non-blanching rash, photophobia	Meningitis, Meningoencephalitis
Ear, Nose, Throat	Earache, muffled hearing, Purulent nasal discharge, facial pain, headache, Sore throat, cervical lymphadenopathy, hoarse voice	Otitis media, sinusitis, pharyngitis
Respiratory Tract	Cough with sputum most likely bacterial, hypoxia, pleuritic pain, chest pain, sore throat	Pneumonia (viral & bacterial)
Cardiac	Regurgitant murmur, embolic lesions, Splinter haemorrhages and Janeway lesions	Infective endocarditis
Breast	Mastalgia, breast engorgement, fever and rigors. Inflamed breast, pain, features of sepsis and 'bruising' suggests deeper infection such as necrotizing fasciitis	Mastitis may lead to necrotising fasciitis, toxic shock syndrome
Hepatobiliary	Right upper quadrant pain, Murphy sign positive for cholecystitis	Cholecystitis, Cholangitis
Abdomen and Pelvis	Abdominal pain, nausea and vomiting, diarrhoea, McBurney's sign for appendicitis, atypical presentations common in pregnancy, acute abdomen	Pelvic collection, peritonitis, appendicitis, pancreatitis, cholecystitis
Urinary Tract	Dysuria, increased urinary frequency, loin pain, acute urinary retention	Cystitis, pyelonephritis, Covert genital herpes
Genital Tract	Offensive vaginal discharge/lochia/liquor, abdominal pain, heavy vaginal bleeding	Chorioamnionitis, endometritis. Think Group A Strep
Wound Infection	Wound erythema/discharge, fever and rigors, severe pain, necrotic tissue Skin and soft-tissue infections are particularly associated with early toxic shock syndromes. (Rash and conjunctival redness)	Cellulitis, necrotising fasciitis, peripheral cannula thrombophlebitis, caesarean birth or episiotomy wounds
Necrotising Fasciitis	Cardinal feature is of agonising pain, typically necessitating increasing amounts of analgesia culminating in opioids, there may be no skin changes early on Later, ascending infection to the skin, producing blisters and frank necrosis	Early necrotising fasciitis (deep in tissues)
Regional anaesthetic blockade-related infection	Headache, photophobia, back pain	Meningitis, spinal abscess

3.3 Assessment

If a pregnant or postpartum patient presents in the community or in hospital feeling unwell think “COULD THIS BE SEPSIS”. Check their observations and refer for medical assessment

Do a general history and examination to try to identify the source of sepsis. The natural adaptations to the body with pregnancy may mask the signs and symptoms of infection or an acute abdomen until the women deteriorates. Specific symptoms and possible diagnoses to consider are outlined in section 3.2. However, symptoms may be less distinctive such as feeling generally unwell; therefore, a high index of suspicion is necessary. Consider and investigate other non-infective causes that can masquerade as sepsis such as pre-eclampsia or other rare diseases (e.g. thrombotic thrombocytopenic purpura, misoprostol side-effects, acute fatty liver of pregnancy). The possibility of COVID-19 must also be considered, and an assessment should be done as per the recommendations.

Ask the following questions

- Has the woman been in contact with someone who has been unwell in the last 7 to 21 days?
- Has the woman been abroad?
- Is the woman feeling unwell?

General symptoms

- **Woman who feels unwell**
- Fever: persistent spikes of temperature may suggest an abscess. Fever may be absent with the use of Paracetamol or of non-steroidal anti-inflammatory drugs (NSAIDS) used post-delivery.
- Rigors, Myalgia
- Diarrhoea or vomiting
- Rash (generalised maculopapular rash)
- Genital tract sepsis may present with constant severe abdominal pain and tenderness not relieved by usual analgesia
- Severe infections may be associated with preterm labour
- Viral infections such as disseminated HSV is a rare and potentially fatal disease if contracted in the peripartum period. While it can present with encephalitis, or rarely febrile hepatitis, in many cases there are no obvious clinical signs of disseminated herpetic infection which can differentiate it from a bacterial cause of sepsis. Pregnancy is a risk factor for HSV hepatitis, which is responsible for 2%–4% of all acute hepatitides. The sicker the woman is with disseminated HSV infection (in the absence of rash), the more likely the incorrect diagnosis of HELLP or AFLP.

Signs

- **Temperature** (pyrexia of ≥38°C or 37.5-37.9°C on 2 consecutive occasions 1 hour apart or hypothermia <36°C)
- **Tachycardia** (>100 bpm)
- **Tachypnoea** (>20 breaths per minute)
- Hypoxia with Oxygen saturations ≤92% without oxygen therapy
- Hypotension with systolic blood pressure (BP) <90mmHg
- Poor peripheral perfusion (capillary refill >2 seconds)
- Oliguria (<0.5ml/kg/hour)
- Impaired mental state

- Worsening clinical condition despite treatment
- Bruising or discoloration of skin (suggests late fasciitis), non-blanching rash

3.4. Management of Suspected Sepsis

Use the Sepsis Screening Tool Acute Assessment and Sepsis 6 pathway (Appendix 1) which encompasses the points below:

1. **Ensure Senior Clinician attends**
    - Inform senior obstetrician and anaesthetist. Ensure senior clinical review within 1hr.

2. **ABC and Oxygen therapy**:
    - Assess airway, breathing and circulation.
    - Start oxygen if oxygen saturations (O2 sats) ≤92%. Aim for target O2 sats of 94-98%. If at risk of hypercarbia aim for O2 sats of 88-92%. Any patient who is shocked or unconscious should immediately receive high flow oxygen at 15 L/min.

3. **Obtain IV access and take bloods:**
    - **Microbiology investigations** – see section 3.5 – ideally obtain samples including blood cultures prior to starting antibiotics, review results as soon as available to allow optimisation of antibiotic regime. Consider HSV polymerase chain reaction (PCR) when indicated
    - **Blood investigations** – see section 3.5
    - **Arterial and venous blood gas** to exclude metabolic acidosis and check lactate level should be measured urgently in women with features making them at high risk of sepsis. Serum lactate of 4 mmol/l or more should prompt immediate escalation of care, including consideration of discussion with the critical care team. A lactate level ≥ 2mmol/l indicates tissue hypoperfusion. Note that lactate measurement may be transiently elevated during and immediately after normal labour and birth. If unsure repeat sample. Ensure all air is excluded from blood gas syringe immediately after taking sample

4. **Give IV Antibiotics within 1hr**
    - The Golden Hour – ideally give after microbiology samples taken. See the trust Antibiotic Guidelines for Obstetrics & Gynaecology (2020) and consider allergy status. Antivirals may also be required.
    - Review antibiotics at 24-48 hours and if there is no response to treatment or the condition is deteriorating at any time arrange prompt patient review by on-call obstetric consultant and involve a **consultant microbiologist** for antibiotic guidance.
    - With any history of MRSA in a woman with sepsis, the addition of vancomycin is advisable.
    - If a change in antibiotic regimen is considered because of deterioration despite first-line treatment, consider the addition of IV Acyclovir 500mg 8 hourly as part of the second line regimen

5. **IV fluids and strict fluid balance documented on a fluid balance chart**
    - Over and under replacement are both potentially serious.
    - Start IV fluids. Give a fluid bolus of 500mls of Plasmalyte or 0.9% sodium chloride if age >16 or 20ml/kg if age <16. NICE recommends using lactate to

guide further fluid therapy. If the initial lactate was >2mmol/L consider repeat measurements at least every hour until it has normalised.
- Consider urinary catheter to measure hourly output – call doctor to review if urine output <0.5ml/kg/hour

6. **Monitor on MEWS chart**–

Temperature, pulse, blood pressure, oxygen saturations and respiratory rate must be measured and documented 4 hourly, as a minimum. Planned frequency of observations must be documented in the notes. Abnormal findings as per MEWS pathway must ALWAYS be escalated IMMEDIATELY to the on call obstetric doctor

Further Management

7. **Paracetamol 1g IV, orally or rectally –** route of administration depends on the clinical situation and is weight dependent

8. Electronic **fetal heart** monitoring – record and consider delivery timing and options if the woman has not delivered.

9. Consider venous thromboembolism **(VTE) prophylaxis**

10. Adequate **pain management** is vitally important in patents affected by sepsis, it aids in reducing anxiety, stress and fear, consider IV diamorphine. Use the pain score on the MEWS chart to monitor progress

11. Blood **Glucose** control: aim for blood glucose 6-10mmol/L if the patient has diabetes (pre-existing or gestational)

12. Involve **Multidisciplinary team** of:
    - Midwife coordinator
    - Senior obstetrician
    - Senior obstetric anaesthetist
    - Microbiologist
    - Intensive care team: in severe/rapidly deteriorating cases consider electrocardiogram, central venous pressure (CVP) and arterial lines.
    - Consider informing the on call clinical site practitioner

13. **Rescreening:**
    - Any patient who is deteriorating or not improving clinically should have the sepsis observations reviewed and investigations repeated if required.

14. **Care plan**:
    - A clear plan should be made in conjunction with the patient and relatives. That plan should be reviewed regularly by a senior clinician, taking into account the whole clinical picture, including response to treatment.

15. There should be an urgent referral to the critical care team in severe or rapidly deteriorating cases of sepsis where the facilities/skills to care for a woman are not available on the maternity unit.

16. During and after pregnancy it should be noted that current systemic infection is an important risk factor for venous thromboembolism and should prompt reassessment

for the correct thromboprophylaxis according to current guidelines

17. Source control should be a priority, e.g. surgery to drain pus, expedited birth where necessary. Women suspected of having infected retained products of conception should be given antibiotics and surgery to remove the infected products of conception should be performed promptly.

3.5. Management of Pyrexia in Labour

In established labour women have their temperature monitored hourly. If this is raised think 'could this be sepsis' and review the signs and symptoms in section 3.3. If sepsis is suspected manage as in section 3.4.

Consider paracetamol for women in labour with a raised temperature of 38 °C or 2 temperatures 37.5–37.9°C 1 hour apart. Be aware that paracetamol is not a suitable treatment for sepsis and it may mask worsening fever, and its use should not delay investigation and treatment when sepsis is suspected.

The normal physiological changes in labour or having an epidural can lead to pyrexia, tachycardia and leucocytosis in the absence of infection. However, treatment with IV antibiotics should still be started if there is suspicion of infection, in order to ensure early treatment is not missed in the small number of women who actually have sepsis in labour.

3.6. Investigations for Sepsis or Suspected Sepsis

Venous/arterial blood samples	**Microbiology**
Full blood count (FBC)/ urea and electrolytes (U&E) / Liver Function tests (LFTs)	Appropriate virological testing, including SARS-Cov-2 testing should be undertaken if indicated
Clotting	Genital Tract swab, wound swab
C Reactive Protein (CRP) PCT (procalcitonin levels)	Products of conception if applicable
Group & Save (G&S) and crossmatch	Urine cultures midstream urine (MSU) / catheter specimen urine (CSU)
Blood cultures (each set including an aerobic and anerobic bottle)	Cerebro-Spinal Fluid (CSF) if appropriate
Blood gases / serum lactate	Other swabs to consider: Stool, sputum, throat, Influenza, placental,perinealand peripheral/central line swab
Blood glucose	
	If the MRSA status of the woman is unknown, a premoistened nose swab should be sent for rapid MRSA screening where such testing is available

- Any woman with symptoms of tonsillitis should have a **throat swab** sent for culture. If there is a history of recent sore throat or prolonged (household) contact with family members with known streptococcal infections (pharyngitis, impetigo and cellulitis) have been implicated in numerous cases of puerperal GAS sepsis.

- An H1N1 swab (nasal viral transport medium swab) should be sent if influenza is suspected clinically. Treatment should be started, and the patient should be barrier nursed until results are obtained (see up-to-date DoH or RCOG guideline on internet). Influenza related morbidity and mortality is associated with a body mass index (BMI) of over 30 kg/m2. Early birth may be indicated during the third trimester of pregnancy. Pulmonary haemorrhage due to primary influenza has been reported. Secondary bacterial infection with pneumococci, beta-haemolytic streptococci and S. aureus occur typically after influenza and other viral respiratory infections.

- If there is a history of recent foreign travel to a tropical destination consider sending investigations for malaria (3 malaria blood films or rapid antigen diagnostic tests over 2 days), dengue fever, enteric fever (typhoid or paratyphoid), HIV seroconversion and rickettsia infection.

- Diarrhoea warrants routine stool culture (e.g. Salmonella, Campylobacter) as well as testing for Clostridium difficile toxin, the latter especially if diarrhoea follows antimicrobial therapy.
- Laboratory must be informed if there is a clinical indication for investigations for unusual pathogens such as Listeria monocytogenes, hospitalisation (multi-drug-resistant organism screening) or there is a history of foreign travel (parasites, malaria, typhoid, cholera, viral haemorrhagic fever, brucella).

- In suspected HSV cutaneous infection, or vulval/sacral pain/paraesthesia with no visible lesion, send a viral swab for PCR either from the lesion, or as a vaginal sample. If disseminated HSV and hepatitis is suspected, send an EDTA blood sample for HSV viral PCR. Discuss management with an infection specialist

Laboratory results suggestive of sepsis:

- Raised white cell count (WCC)
- Low WCC (lymphopaenia in toxic shock)
- Thrombocytopaenia +/- abnormal coagulation
- Falling haemoglobin (with little relative decrease in haematocrit) suggests intravascular haemolysis typical of Streptococcal infection
- Raised C-reactive protein
- Raised blood glucose
- Raised creatinine
- Abnormal liver function tests - raised bilirubin, falling albumin. If LFTs show an anicteric pattern (i.e. markedly raised ALT with normal bilirubin) consider a diagnosis of herpetic hepatitis and treat appropriately, while waiting for results to exclude other causes of hepatitis

- Metabolic acidosis +/- raised lactate
- Rising creatine kinase - suggestive of myositis/necrotising fasciitis
- Raised white cells in urine with a nitrite positive dipstick (MSU only)
- Positive blood cultures / swab result
- Thrombocytosis (high platelet count) with a rising CRP and a swinging pyrexia often indicates a collection of pus or an infected haematoma

Imaging

Prompt imaging may identify the source of infection.

- Chest x-ray (CXR)
- Pelvic ultrasound (USS)
- Computed tomography (CT) if pelvic abscess is suspected
- Magnetic resonance imaging (MRI)

3.7. Antibiotics

Broad-spectrum antibiotics within 1 hour of recognition of severe sepsis with or without septic shock must be administered. If you are confident about the source of infection, then the antibiotic choice should be tailored to cover the likely pathogens according to hospital policy. If you are unsure of the source then a broad spectrum agent covering gram positives and gram negatives should be used with anti-pseudomonal cover. In either situation refer to the Antibiotic Guidelines for Obstetrics & Gynaecology 2025. The choice and need for antibiotic therapy should be reviewed daily, and should be de-escalated based upon culture results, if appropriate, in order to reduce the development of antimicrobial resistance and toxicity.
If the woman has been on broad-spectrum antimicrobials before developing signs of sepsis it is likely the pathogen is multi-drug resistant, and microbiologist should be consulted for advice on further antimicrobial options.

It is important to remember not all sepsis is caused by bacteria and consideration should be given to anti-fungal treatment in patient with certain risk factors (e.g. patients with solid organ transplants or HIV positive patients).

3.8 Indications for Transfer to ICU

Multidisciplinary communication is essential, between the obstetric and anaesthetic consultants on Labour Ward, the Intensive Care Unit (ICU) consultant and the microbiology consultant. If the obstetric consultant is not already present on Labour Ward at the time of consideration for ICU transfer, they must be informed by the obstetric registrar and the obstetric consultant must telephone the other consultants listed above.

SYSTEM	INDICATION
Cardiovascular	Hypotension (< 90 mmHg systolic) or raised serum lactate (> 4 mmol/l) persisting despite fluid resuscitation, suggesting the need for vasopressor and/or inotrope support
Respiratory	Mechanical ventilation Airway protection Pulmonary oedema
Renal	Renal replacement therapy
Neurological	Significantly decreased consciousness level
Miscellaneous	Multi-organ failure Uncorrected acidosis Hypothermia

3.9. Fetal Monitoring and Delivery

During the intrapartum period, continuous electronic foetal monitoring (CTG) is recommended for gestations ≥28+0 weeks in the presence of maternal pyrexia. Objective evidence of in-utero infection is associated with an abnormal CTG; however, it is not a sensitive predictor of foetal in-utero systemic inflammatory response or early onset neonatal sepsis.

Changes in the CTG such as changes in baseline variability or new onset decelerations should prompt reassessment of the maternal observations and progress in labour.

The decision on mode of delivery should be individualised with consideration of severity of maternal illness, duration of labour, gestational age and viability. Epidural or spinal anaesthesia should be discussed with the anaesthetic consultant on call; a general anaesthetic may be required for caesarean section.

The risk of neonatal encephalopathy and long-term cerebral palsy is increased in the presence of in utero infection.

3.10 Neonatal Care

A woman with suspected or confirmed maternal sepsis or possible chorioamnionitis, the on call neonatal doctor must be called after the delivery to assess the need for blood cultures and antibiotics for the neonate.

3.11 Sepsis Prevention

Antenatal

- **Influenza vaccination:** this has been shown to reduce the risk of flu-associated acute respiratory infection in pregnant women by one half
- **Antenatal counselling and advise to at risk women:** women should be advised to contact their GP or maternity triage immediately if they feel unwell, are experiencing symptoms of sepsis or in circumstances where they are more prone to sepsis such as premature rupture of membranes.
- **Prophylactic antibiotics:** this may be indicated for at- risk women including those with PPROM, recurrent UTIs and those undergoing cervical cerclage insertion.
- **Group A Streptococcus (GAS):** any GAS identified in pregnancy (e.g. on throat swab) should be treated to avoid invasive GAS infection.

Intrapartum

- **Group B Streptococcus prophylactic antibiotics:** refer to antibiotic guideline for obstetrics
- **Vaginal cleansing prior to Caesarean section:** all patients should receive intra-vaginal aseptic preparation with iodine / chlorhexidine prior to commencing the procedure.
- **Antibiotic prophylaxis for instrumental deliveries:**
- Refer to Trust’s antibiotic guideline for obstetrics

Postnatal

- **Wound and Perineal care advice:** daily cleaning of the perineum, regularly changing maternity pads and good hand hygiene can reduce the risk of genital tract sepsis.
- **Prophylactic antibiotics** for 3rd/4th degree tears, manual removal of placenta and intrauterine balloon insertion.
- **Good communication amongst healthcare workers** to facilitate good discharge planning and handover to the community care team (district midwife and GP) so that women who have required treatment for sepsis in hospital can have appropriate follow up visits arranged. All community midwives must carry a thermometer so that they can check postnatal maternal temperature.

- The 2023 MBBRACE report advises that streptococcal infections, particularly Group A Streptococcus (GAS), often present within 12 hours after childbirth. Any woman with significant abdominal pain or ongoing vaginal bleeding post-delivery should be assessed for infection. Persistent pain after vaginal birth that does not respond to simple analgesia warrants prompt face-to-face evaluation, including abdominal and vaginal examination if needed. If there are concerns or repeated visits to the GP or emergency services, the woman should be referred immediately to the obstetric unit or A&E.
- GAS can cause endometritis, myonecrosis, peritonitis and streptococcal toxic shock syndrome (STSS)

3.12 Infection Control Issues

MRSA and Group A Streptococci are easily transmitted via the hands of health care workers and via close contact in households. Local Infection control guidelines should be followed for hospital-specific isolation and contact precautions. See “policy for the control of methicillin resistance staphylococcus aureus (MRSA). This currently includes MRSA swabs for:

- In-utero transfer admitted from another hospital or has been at another hospital
- Woman who has had MRSA before and is having an elective caesarean section

Local advice from the Infection Control Team should always be sought.

EARLY INVOLVEMENT OF THE MULTIDISIPLINARY TEAM IS MANDATORY.

USE OF UNIVERSAL ASEPTIC TECHNIQUES AND HANDWASHING IS MANDATORY.

4. Associated Guidelines, Polices and Documents

Hand Hygiene Policy
MRSA Control (Methicillin Resistant Staphylococcus Aureus)
Meningitis Including Meningococcal Septicaemia
Isolation Policy (Infection Control)
Risks Associated with Infection Prevention and Control
Standard Infection Control Precautions
Urinary Catheter Policy
Venous Thromboembolism Prophylaxis in Pregnancy and the Puerperium
Maternity Early Warning Scoring (MEWS) - Early Recognition of the Severely Ill Pregnant Woman

Maternal Protocol for Prevention of Early Onset Neonatal Group B Streptococcal Disease
Management of Suspected Neonatal Infection

5. Training need identified

Training for this guideline is provided as described in the Trust Training Needs Analysis which will include an annual update on infection control.

6. Implementation and Monitoring

All guidelines, protocols and procedures are available on the intranet. All members of staff have a responsibility to check the Trust's intranet and ensure that they are referring to the most recent issue of the document. The weekly general information bulletin includes notification of publication of new/updated documents.

This clinical guideline, protocol or procedure will be monitored using local and national guidance and via the Trust Incident and Risk Management Policies. If risks/issues are identified further specific monitoring will be considered, for example, clinical audit or risk assessment.

7. References

- National Institute for Health and Care Excellence (2017) (NG51) Sepsis: recognition, diagnosis and early management.
- National Institute for Health and Care Excellence Intrapartum care for women with existing medical conditions or obstetric complications and their babies (2019) (NG121)
- Sepsis in Pregnancy, Bacterial RCOG Green top guideline 64a April 2012
- Sepsis following Pregnancy, Bacterial RCOG Green top guideline 64b April 2012
- MBBRACE 2019 Saving Lives, Improving Mothers' Care. Lessons learned to inform maternity care from the UK and Ireland Confidential Enquiries into Maternal Deaths and Morbidity 2015-2017
- The UK Sepsis Trust Screening and action tool for use in pregnancy 2020. https://sepsistrust.org/professional-resources/clinical-tools/
- The Sepsis manual 5th edition from the UK Sepsis Trust 2019. https://sepsistrust.org/wp-content/uploads/2020/01/5th-Edition-manual-080120.pdf
- MBRRACE- UK, "Saving Lives Improving Mothers' Care—Lessons Learned to Inform Maternity Care From the UK and IrelandConfidential Enquiries Into Maternal Deaths and Morbidity 2019–21"(2023).
- Identification and Management of Maternal Sepsis During and Following Pregnancy Green-top Guideline No. 64

8. Appendices

Appendix 1 - Modified from Sepsis 6 pathway

Appendix 1

The Hillingdon Hospitals NHS Foundation Trust

SEPSIS SCREENING TOOL ACUTE ASSESSMENT

PREGNANT OR UP TO 6 WEEKS POST-PREGNANCY

PATIENT DETAILS:

DATE: **TIME:**
NAME:
DESIGNATION:
SIGNATURE:

01 START THIS CHART IF THE PATIENT LOOKS UNWELL OR MEOWS HAS TRIGGERED

RISK FACTORS FOR SEPSIS INCLUDE:

- ☐ Impaired immunity (e.g. diabetes, steroids, chemotherapy)
- ☐ Recent trauma / surgery / invasive procedure
- ☐ Indwelling lines / IVDU / broken skin

02 COULD THIS BE DUE TO AN INFECTION?

YES

LIKELY SOURCE:

- ☐ Respiratory
- ☐ Breast abscess
- ☐ Urine
- ☐ Abdominal pain / distension
- ☐ Infected caesarean / perineal wound
- ☐ Chorioamnionitis / endometritis

NO → **SEPSIS UNLIKELY, CONSIDER OTHER DIAGNOSIS**

03 ANY RED FLAG PRESENT?

YES

- ☐ Objective evidence of new or altered mental state
- ☐ Systolic BP ≤ 90 mmHg (or drop of >40 from normal)
- ☐ Heart rate ≥ 130 per minute
- ☐ Respiratory rate ≥ 25 per minute
- ☐ Needs O_2 to keep SpO_2 ≥ 92%
- ☐ Non-blanching rash / mottled / ashen / cyanotic
- ☐ Lactate ≥ 2 mmol/l*
- ☐ Not passed urine in 18 hours (<0.5ml/kg/hr if catheterised)

*lactate may be raised in & immediately after normal delivery

YES → **RED FLAG SEPSIS**

START **SEPSIS SIX**

04 ANY AMBER FLAG PRESENT?

NO

- ☐ Acute deterioration in functional ability
- ☐ Respiratory rate 21-24
- ☐ Heart rate 100-129 or new dysrhythmia
- ☐ Systolic BP 91-100 mmHg
- ☐ Has had invasive procedure in last 6 weeks (e.g. CS, forceps delivery, ERPC, cerclage, CVs, miscarriage, termination)
- ☐ Temperature < 36°C
- ☐ Has diabetes or gestational diabetes
- ☐ Close contact with GAS
- ☐ Prolonged rupture of membranes
- ☐ Bleeding / wound infection
- ☐ Offensive vaginal discharge
- ☐ Non-reassuring CTG / fetal tachycardia >160
- ☐ Behavioural / mental status change

YES →

FURTHER REVIEW REQUIRED:

- SEND BLOODS AND REVIEW RESULTS
- ENSURE SENIOR CLINICAL REVIEW within 1HR

TIME OF REVIEW: ☐☐:☐☐

ANTIBIOTICS REQUIRED:

☐ Yes ☐ No

NO AMBER FLAGS = ROUTINE CARE /CONSIDER OTHER DIAGNOSIS

THE UK SEPSIS TRUST

UKST 2020 1.4 PAGE 1 OF 2 / UKST, REGISTERED CHARITY 1158843

SEPSIS SCREENING TOOL - THE SEPSIS SIX

PREGNANT
OR UP TO 6 WEEKS POST-PREGNANCY

PATIENT DETAILS:	DATE:	TIME:
	NAME:	
	DESIGNATION:	
	SIGNATURE:	

COMPLETE ALL ACTIONS WITHIN ONE HOUR

01 ENSURE SENIOR CLINICIAN ATTENDS

NOT ALL PATIENTS WITH RED FLAGS WILL NEED THE 'SEPSIS 6' URGENTLY. A SENIOR DECISION MAKER MAY SEEK ALTERNATIVE DIAGNOSES/ DE-ESCALATE CARE. RECORD DECISIONS BELOW

NAME: GRADE:

TIME ☐☐:☐☐

02 OXYGEN IF REQUIRED

START IF O_2 SATURATIONS LESS THAN 92% - AIM FOR O_2 SATURATIONS OF 94-98%
IF AT RISK OF HYPERCARBIA AIM FOR SATURATIONS OF 88-92%

TIME ☐☐:☐☐

03 OBTAIN IV ACCESS, TAKE BLOODS

BLOOD CULTURES, BLOOD GLUCOSE, LACTATE, FBC, U&Es, CRP AND CLOTTING
LUMBAR PUNCTURE IF INDICATED

TIME ☐☐:☐☐

04 GIVE IV ANTIBIOTICS

MAXIMUM DOSE BROAD SPECTRUM THERAPY
CONSIDER: LOCAL POLICY / ALLERGY STATUS / ANTIVIRALS

TIME ☐☐:☐☐

05 GIVE IV FLUIDS

GIVE FLUID BOLUS OF 20 ml/kg if age <16, 500ml if 16+
NICE RECOMMENDS USING LACTATE TO GUIDE FURTHER FLUID THERAPY

TIME ☐☐:☐☐

06 MONITOR

USE MEOWS. MEASURE URINARY OUTPUT: THIS MAY REQUIRE A URINARY CATHETER REPEAT LACTATE AT LEAST ONCE PER HOUR IF INITIAL LACTATE ELEVATED OR IF CLINICAL CONDITION CHANGES

TIME ☐☐:☐☐

RED FLAGS AFTER ONE HOUR – ESCALATE TO CONSULTANT NOW

RECORD ADDITIONAL NOTES HERE:

e.g. allergy status, arrival of specialist teams, de-escalation of care, delayed antimicrobial decision making, variance

THE UK SEPSIS TRUST

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Auditable Topics

Proportion of women with suspected sepsis who receive ap-propriate antibiotic treatment within 1 hour of diagnosis (100%)

Proportion of women with suspected sepsis who have blood cultures obtained prior to initiation of antibiotic treatment. (100%)

Proportion of women with suspected sepsis who have a venous lactate sample obtained within 1 hour of diagnosis. (100%)

Whether a history of previous multi-drug-resistant organisms was checked and influenced the choice of empirical antimicrobials on presentation of maternal sepsis. (100%)

Proportion of women with GAS identified who are appropriately treated and in whom appropriate contact tracing of household contacts and exposed healthcare workers was carried out. (100%)